

2.2 Pre-processing : RegArima modelling in Tramo and in X12/X13

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Outline

- Why pre-processing (with RegArima)
- RegArima model
 - Definition, estimation
- Automatic model identification (AMI)
 - Overview, seasonality tests, log/level, calendar effects, outliers, ARIMA models, final checks
- Final remarks

1. Why pre-processing?

- Exercise:
 - Effects of log/level, trading days, outliers on the final SA series (trend, irregular...)
 - Impacts using Tramo-Seats and X12-Arima

1.2. Why RegArima?

- Flexible and parsimonious
- Efficient estimation procedures
- Historical reasons
 - Link with SEATS and the canonical decomposition
- More powerful alternatives could now be considered (structural models...)

2.1 RegArima model

- *Additive case:*

$$\Delta(B)\Phi(B)(y_c - X\beta) = \Theta(B)\varepsilon$$

- *Multiplicative case:*

$$\Delta(B)\Phi(B)(\ln y_c - X\beta) = \Theta(B)\varepsilon$$

- **Notations:**

- y_c : series corrected for any pre-specified effect.
- X : any regression variable (trend constant, calendar, outliers...)
- $\Delta(B), \Phi(B), \Theta(B)$: differencing, auto-regressive and moving average polynomials

2.2 RegArima estimation (I)

- Exact estimation

$$\Phi(B)(\Delta(B)y_c - \Delta(B)X\beta) = \Theta(B)\varepsilon$$

$$\Phi(B)(\tilde{y}_c - \tilde{X}\beta) = \Theta(B)\varepsilon$$

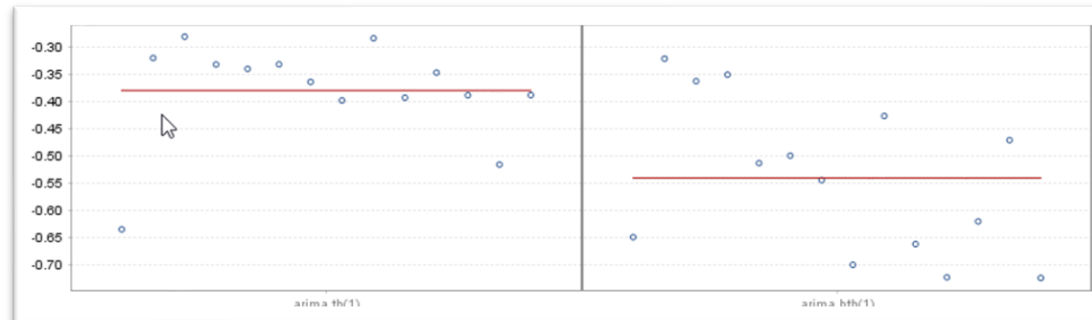
$$\tilde{y}_c = \tilde{X}\beta + \xi, \quad \xi \sim N(0, \sigma^2 \Omega)$$

$$L^{-1}\tilde{y}_c = L^{-1}\tilde{X}\beta + \varepsilon, \quad \varepsilon \sim N(0, \sigma^2 I), \quad LL' = \Omega$$

- Differencing $\rightarrow$ Kalman filter $\rightarrow$ OLS solved by a QR decomposition.
- ML estimation of φ, θ by Levenberg-Marquardt (β, σ^2 concentrated out of the likelihood)
- Residuals 
 - « QR-residuals » (depends on the QR decomposition)
 - Full residuals = $L^{-1}(\tilde{y}_c - \tilde{X}\hat{\beta})$

2.2 RegArima estimation (II)

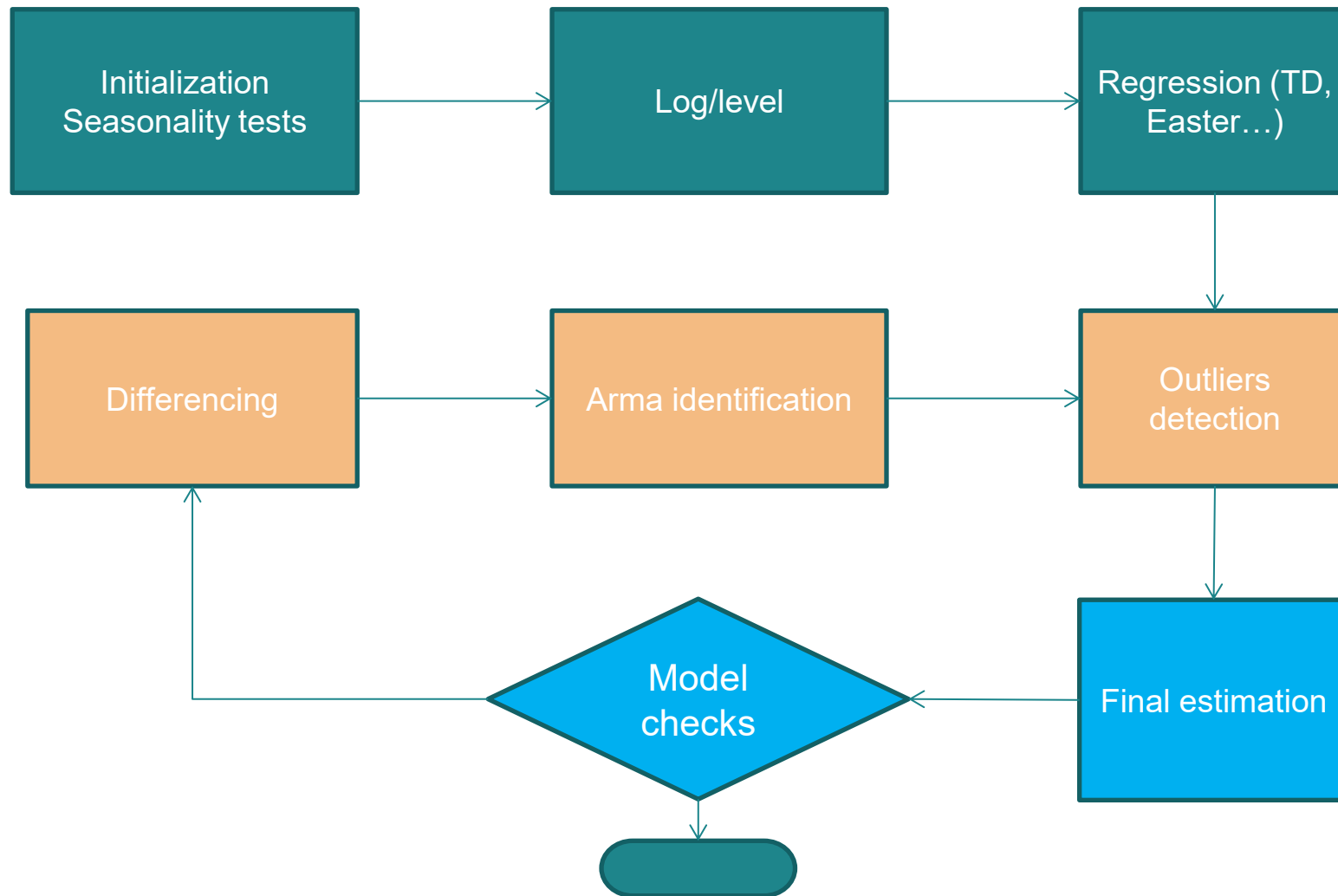
- Fast estimation
 - Differencing $\rightarrow$ OLS (QR decomposition) $\rightarrow$ Residuals (e_t)
 - Estimation of φ, θ by Hannan-Rissanen
 - $e_t = \sum_{i=1}^n e_{t-i} + a_t$
 - $e_t = -\sum_{i=1}^p \varphi_i e_{t-i} + \sum_{i=1}^q \theta_i a_{t-i} + \epsilon_t$
 - Not a real problem: why trying to estimate exact parameters when we know that they are unstable ?
 - Impact of the parameters on the regression coefficients, on the seasonally adjusted series ?



2.2 RegArima estimation (III)

- X12-Arima
 - Exact ML estimation
- Tramo
 - Fast estimation in most intermediary steps
 - Exact ML estimation in final steps

3.1 AML: simplified schema



3.2 Seasonality tests

- Only in the last versions of Tramo and in JD+
- Initial tests
 - Ljung-Box test: auto-correlations at seasonal lags
 - Correlations between $(y(t), y(t-s))$, $(y(t), y(t-2*s))$
 - Friedman (non parametric) test

3.3 Log-level test (I)

Tramo



$$SsqErr \propto likelihood$$

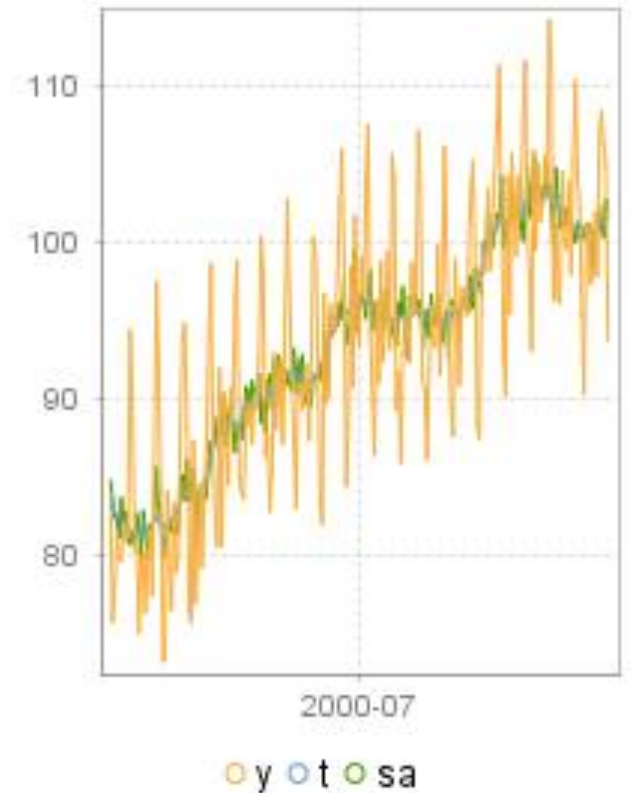
X13



$$AICC = -2 \cdot (\loglikelihood - n \cdot np / (n - np - 1))$$

3.3 Log-level test (II)

- Tramo
 - Levels: mean + airline
 - $Sslevels = 7.736395$
 - **Logs**: mean + airline
 - $Sslogs \cdot gmean(levels)^2 = 7.768457$
 - $logs - levels = -0.03206 > \log(.95) = -0.05129$
- X13
 - **Levels**: lp + td + mean + airline
 - Loglikelihood = -463.920346919396
 - AICC = 938.1264081245064
 - Logs (+lp adjust) : td + mean + airline
 - Adjusted loglikelihood = -469.5936870686463
 - AICC = 947.3769475970083



3.4 Regression (calendar) tests (I)

- Tramo
 - Legacy code:
 - T-tests on trading days, leap year, Easter variable
 - New tests:
 - F-tests on trading/working days,
 - T-tests on trading days, leap year, Easter variable
- JD+
 - Also Wald tests on trading/working days

```
double fdel = (td1Stats.SsqErr - td6Stats.SsqErr) / (5 * sigma);  
if (fdel > 0) {  
    fstat.setDFNum(5);  
    pdel = fstat.getProbability(fdel, ProbabilityType.Upper);  
}
```

3.4 Regression (calendar) tests (II)

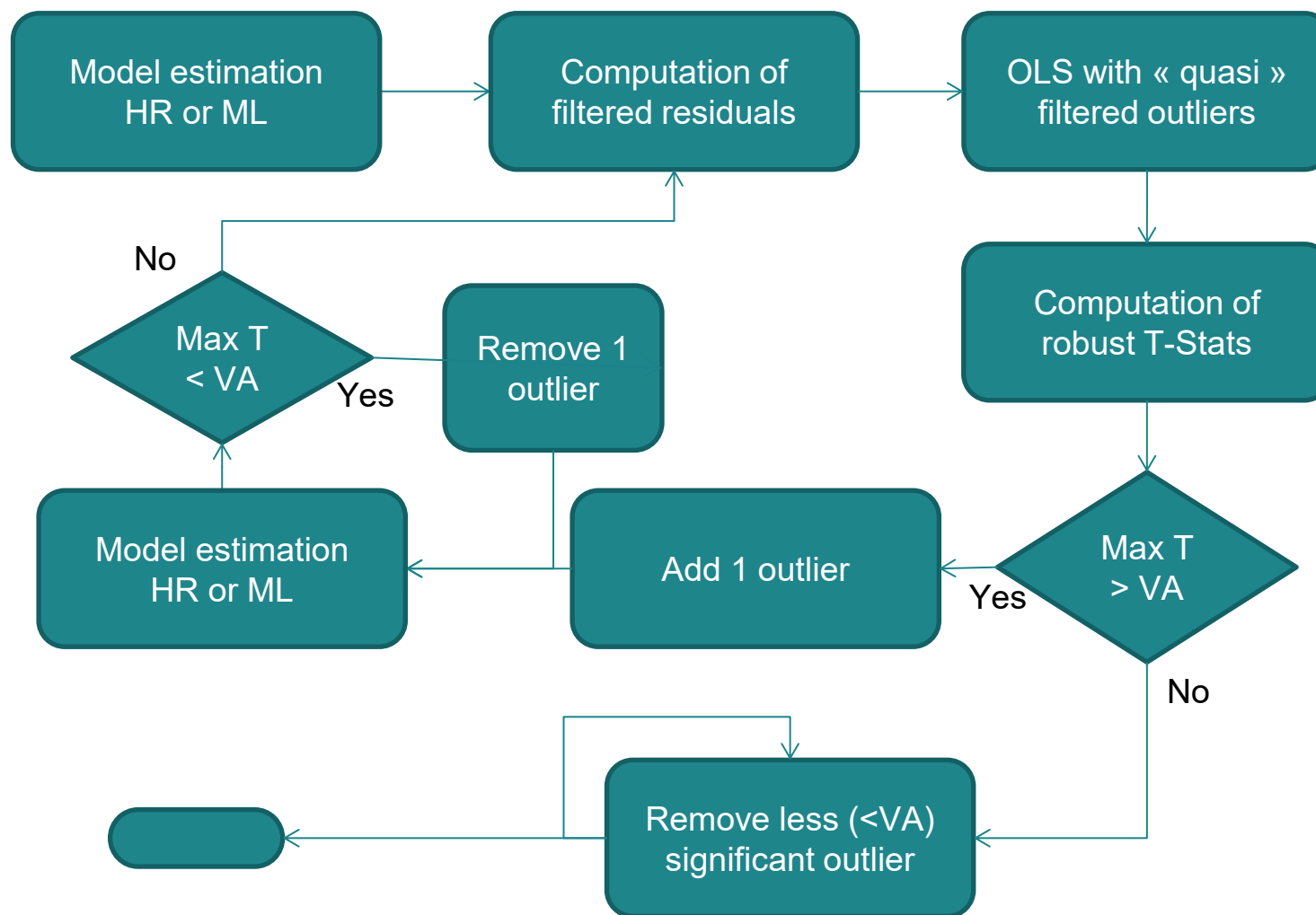
- X12
 - AIC comparison between:
 - Models with and without trading days
 - Models with different Easter variables (length=1, 8, 15)

```
public void calc() {  
    double n = effectiveObservationsCount;  
    double np = estimatedParametersCount;  
    double ll = adjustedLogLikelihood;  
    double nll = logLikelihood;  
    AIC = -2 * (ll - np);  
    HannanQuinn = -2 * (ll - np * Math.log(Math.log(n)));  
    AICC = -2 * (ll - (n * np) / (n - np - 1));  
    BIC = -2 * ll + np * Math.log(n);  
    BIC2 = (-2 * nll + np * Math.log(n)) / n;  
    BICC = Math.log(SsqErr / n) + (np - 1) * Math.log(n) / n; // TRAMO-like  
}
```

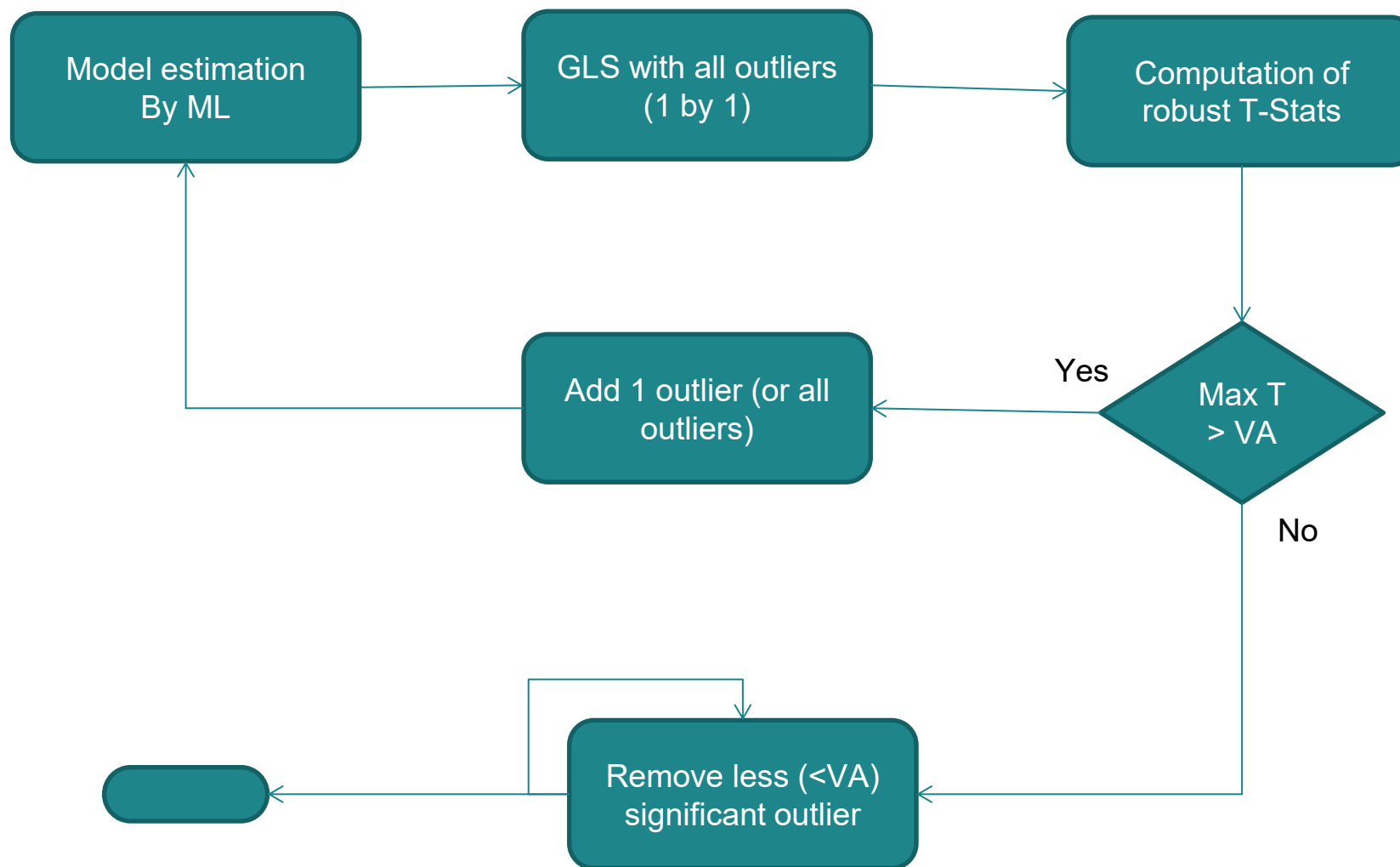
3.5 Outliers detection

- Types of outliers
 - AO, LS, TC, SO (SLS and IO not supported)
- Impact of outliers
 - Estimated parameters
 - Forecasts...
 - Specific SA issues
- Solutions
 - ML-based solutions
 - [Robust estimates]
- Issues
 - Unstable algorithms
 - « Masking » effects

3.5 Outliers detection in Tramo



3.5 Outliers detection in X12



3.5 Outliers detection. Key points

- ML or HR estimation
- Computation of robust stdev on the residuals
- Regression on residuals or on complete models
- Back calculation to identify masking effects
- Exact or approximate GLS estimation



3.61 Differencing

- Tramo ~ X12
- Step 1:
 - Estimate $(2\ 0\ 0)(1\ 0\ 0)+\text{mean}$
 - Select root(s) : $1/|r| > \text{initial UR (0.97)}$
- Step 2:
 - Estimate $(1\ x\ 1)(1\ y\ 1)+\text{mean}$
 - Cancel similar AR, MA roots (cancel)
 - Select root(s) : $1/|r| > \text{final UR (0.91)}$

3.62 ARMA identification

- Tramo ~ X12
- Computes BIC (Tramo-like) for different ARMA orders
 - Estimation of the models:
 - Tramo: Hannan-Rissanen
 - X12: exact ML
- Best (acceptable) model selected

3.7 Final checks

- **Suppression of non-significant ARIMA parameters**
- **Regular/seasonal under-differencing (quasi-unit roots)**
- Seasonality control
- Residual trading days (F-test on the residuals)
- Benchmarking with reference (airline) models
 - Criteria: BIC , outliers, LjungBox, Seasonal LjungBox, Skewness



4. Final (personal) remarks

- Wonderful “expert system”
- But...
 - Several known weaknesses
 - Log/level in case of large outliers
 - Trading days selection (Tramo)
 - Outliers (not robust enough)
 - Selection of non decomposable models
 - ...
- But...
 - Not easy to outperform the current algorithms



4. Final (personal) remarks

- Avoid complex models (too unstable)
- Impact of the ARIMA model on the regression coefficients usually limited
- ARIMA coefficients unstable → pragmatic approach for
 - Time span, revision policies, accuracy of the estimation...
- Fixed regression coefficients (TD) is an annoying limitation → structural models



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